BigQuery architecture

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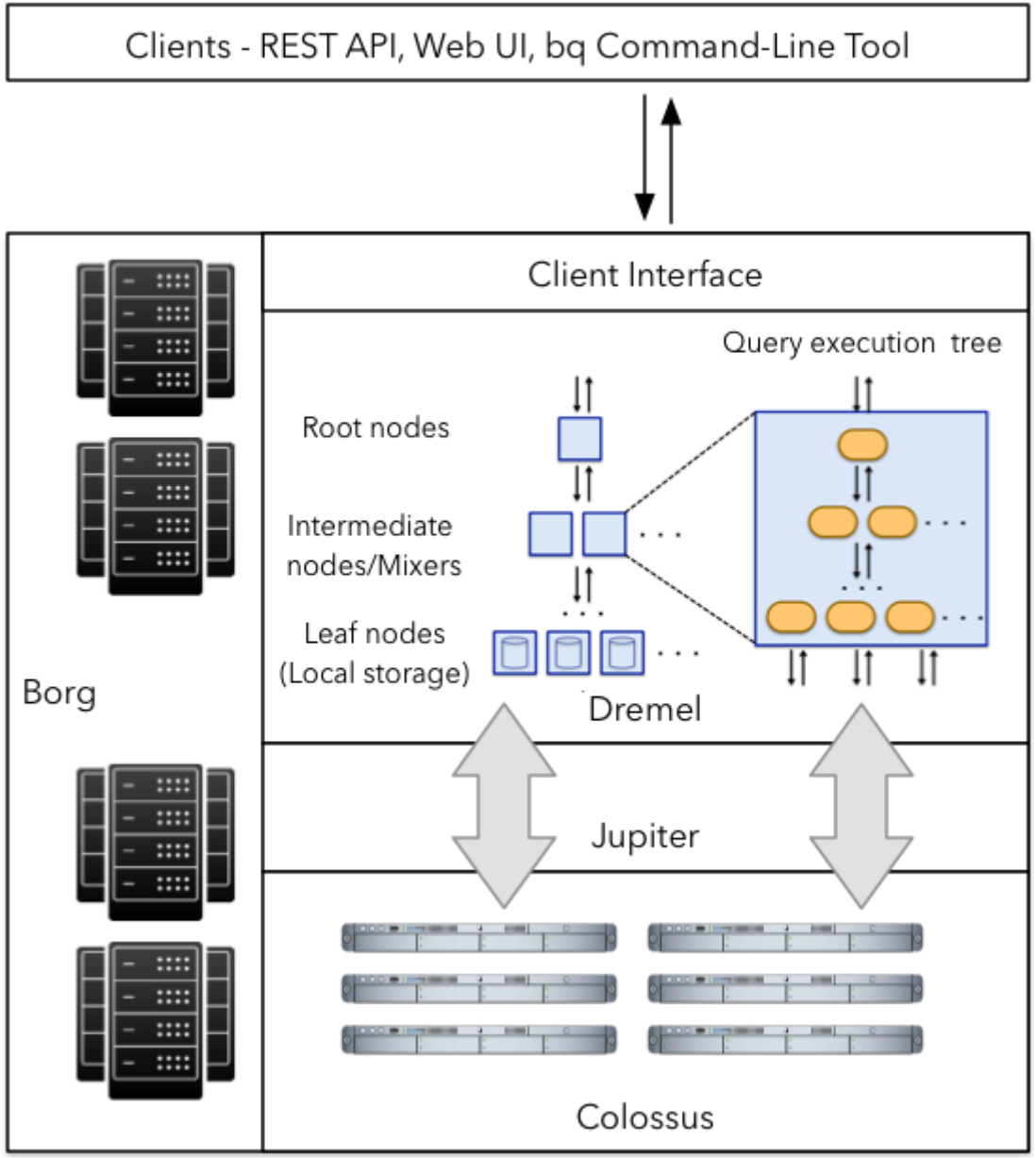
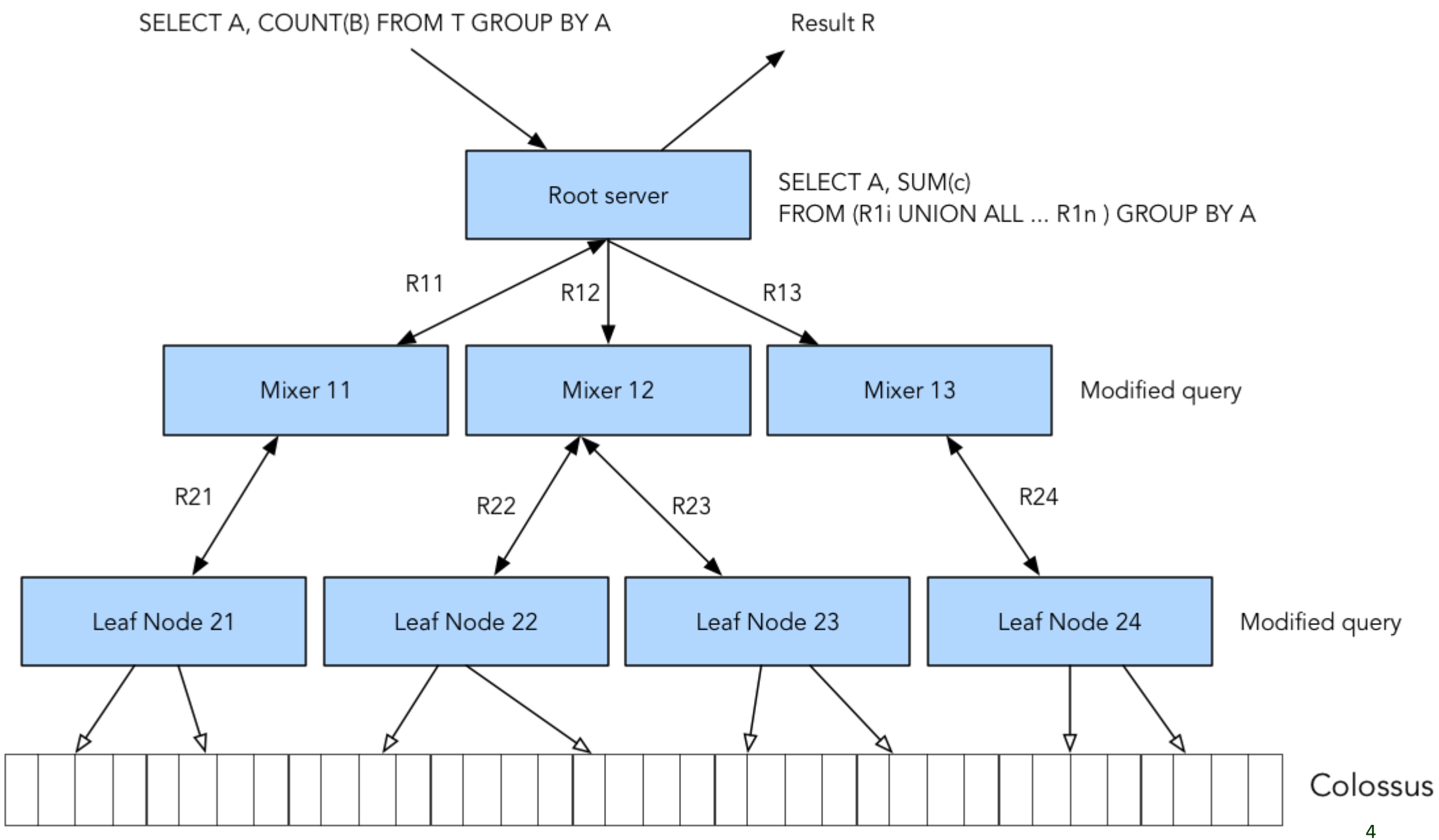
Overview

- Objective is to support read-only queries for decision support
- Compressed column store
- Optional partitioning
- Can be combined with clustering based on tags
- Redundancy/fault tolerance is ensured by the underlying storage endine
- Automatic reorganizaion based on the queries

- Dremel: multi-level query engine -> break down into parallel jobs
- Colossus: independent persistent storage (distributed file system)
- Jupiter: networking (1 petabit/s)
- Borg: compute capacity




SELECT A, COUNT(B) FROM T GROUP BY A
Result R
Root server
SELECT A, SUM(c)
FROM (R1i UNION ALL ... R1n) GROUP BY A
R11
R12
R13
Mixer 11
Mixer 12
Mixer 13
Modified query
R21
R22
R23
R24
Leaf Node 21
Leaf Node 22
Leaf Node 23
Leaf Node 24
Modified query
Colossus